

MACHINE LEARNING WITH COMPLEX DATA STRUCTURES

Time Series, Panel Data, and Clustered Observations

Marco R. Steenbergen 

steenbergen@ipz.uzh.ch

University of Zurich

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Time Series

Time Series Are Different

- So far, we have discussed i.i.d. data—independent and identically distributed.
- Time series are different:
 - Chronological dependence is a key feature of these data.
 - Temporal order cannot be ignore without introducing severe data leakage ([Hewamalage et al. 2023](#)).
- These considerations affect data splitting as well as modeling,

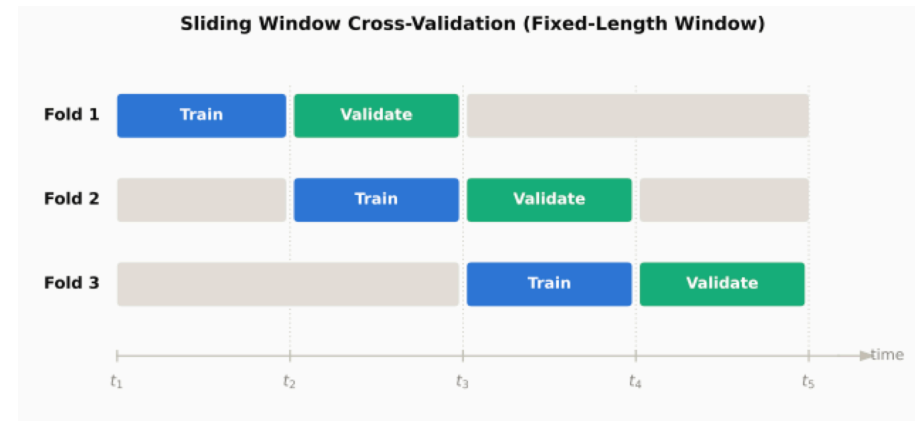
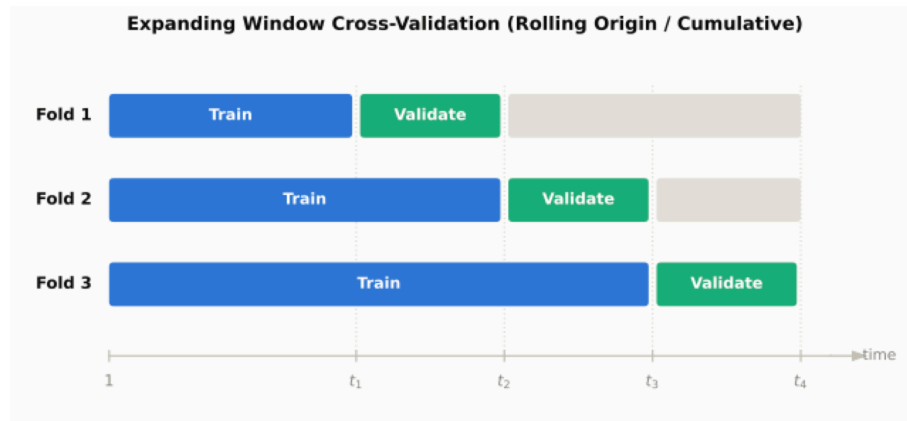
Data Splitting

- Training sets cannot be randomly selected—they must preserve chronology.
- Hence, training data reflect the past and test data the future.
- In `tidymodels`, we use `initial_time_split()`.

Cross-Validation

- Re-sampling for validation must also be handled differently.
- There are two dominant frameworks:
 1. **Expanding window** (a.k.a. rolling CV or cumulative CV)—best when historical patterns over long horizons build stability.
 2. **Sliding window** (a.k.a. rolling window)—best when recent data matters more than distant past (non-stationary environments).

Cross-Validation Cont'd



Cross-Validation Cont'd

Expanding

Sliding

```
1 time_resamples <- rolling_origin(  
2   train_data,  
3   initial      = 12,    # Initial training set size (e.g., 12 periods/months)  
4   assess       = 3,    # Assessment/validation window size  
5   skip         = 1,    # How many steps to skip between folds  
6   cumulative   = TRUE  # TRUE = Expanding window; FALSE = Fixed-length sliding window  
7 )
```

Cross-Validation Cont'd

Expanding

Sliding

```
1 time_resamples <- sliding_period(  
2   train_data,  
3   index      = date_column, # Name of your date/datetime column  
4   period     = "month",    # Split unit: "day", "week", "month", "year", etc.  
5   lookback   = 11,         # Lookback window (e.g., 11 months + current = 12)  
6   assess_stop = 3,         # Assessment window length  
7   step       = 1,          # Move forward by 1 month per fold  
8   complete  = TRUE  
9 )
```


Algorithms

- Feature engineered ML:
 - Tree ensembles do not inherently know sequence; we provide it by engineering lag variables, rolling aggregations, and time components,
- Classical time series econometrics:
 - ARIMA, AutoARIMA, and exponential smoothing.
- Sequence-first neural architectures:
 - RNs and LSTMs.

Repeated Cross-Sections and Panel Data

What Is Different

- We have two dimensions of variation:
 1. Cross-sections that may display clustering.
 2. Time within entities, with autocorrelation over time in the same entity.
- How we split panel data, fundamentally changes the model's learned structure ([Sweet et al. 2023](#)).

Re-Sampling Options

Approach	Mechanism	What It Tests	Implementation
Group/Unit v -fold	Splits by identity ID— all time periods for i stay in the same fold.	Generalization to unseen entities.	<code>group_vfold_cv(data, group = entity_id)</code>
Temporal panel split	Cut at time t_1 across all entities	Generalization to future time points.	<code>initial_time_split()</code> or <code>sliding_period()</code>
Block 2D split	Hold out both unseen entities and future time points	Generalization to unseen entities in the future	Custom grouping on ID + date cutoffs

Algorithms

- Classical algorithms may require feature engineering:
 - Demeaning
 - Entity-level historical averages (Mundlak)
- Some interesting new work on double machine learning for panels (**Clarke and Polsell 2026**).

Hierarchical Data

Re-Sampling

- Cluster-level splits:



Hold out entire higher-level units.



Generalizability to unseen clusters.

- Stratified within-group splits:



Samples individuals from within every group—all clusters are present in test and training sets.



Generalizability to unseen cases within known groups.

Algorithms

- Mixed effects random forests (MERFs) ([Hajjem et al. 2014](#)) or binary mixed models (BiMMs) ([Speiser et al. 2019](#)):

→

$$y_{ij} = f(\mathbf{x}_{ij}, \boldsymbol{\theta}) + \mathbf{z}_{ij}^{\top} \boldsymbol{\beta}_j + \text{error}$$

→

First part is non-parametric; second part is linear.

- Mundlak-trees:
 - Tree-based learners with individual features, means, cluster encodings.
- Tree-boosting with random effects ([Sigrist 2022](#)):
 - Combines mixed effects with gradient boosted decision trees.

References

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